

THARA AI

CEO Assistant | Intelligent Data Analytics Platform

Enterprise-Grade Bilingual Voice-First Analytics Engine
Talk to your data in English or Tamil. Get instant answers.

<2s

Avg Response

9

Data Connectors

29

API Endpoints

155

Test Cases

5

AI Pipeline Layers

16

Tables Auto-Profiled

0

TypeScript `any`

550+

Logs Structured

TECHNICAL ACHIEVEMENTS & USPs

- Multi-Step Cross-Table Query Engine -- joins data across tables without complex SQL
- Self-Healing SQL Executor -- 5+ automatic recovery strategies, 3 retries, 0.90 fuzzy threshold
- Native Tamil + English Bilingual -- not a translation wrapper; native NLP in both languages
- Voice-First UX -- VAD, streaming TTS (200ms TTFB), live word-by-word captions
- Dataset-Agnostic -- works with ANY domain (sales, HR, healthcare, education, logistics)
- Zero Hallucination Guardrails -- data-only responses, no LLM speculation
- Plug-and-Play Data -- 9 connectors, auto-load Google Drive folders, zero schema setup
- Intelligent Routing -- LLM-primary table selection, 60% fewer tokens than top-k approaches
- Production-Hardened -- rate limiting, SSRF prevention, Sentry, CI/CD, 100% pinned deps
- Personality-Driven -- named assistant 'Thara' with 50+ natural conversation patterns

1. SYSTEM ARCHITECTURE

5-Layer Intelligent Query Pipeline

Layer 1: Routing

LLM-primary table selection with rule-based fallback. Confidence scoring (high >60%, low <15%). Context-aware follow-up detection.

Layer 2: Planning

Gemini 2.0 Flash generates structured query plans. Multi-step support for cross-table joins with variable substitution.

Layer 3: Validation

JSON schema validation, date format normalization, column existence checks before execution.

Layer 4: Execution

DuckDB in-memory SQL engine. Self-healing with fuzzy matching (0.90 threshold). 3 retry attempts with 5+ fix strategies.

Layer 5: Explanation

LLM-powered natural language response. Emotion detection, personality-driven, bilingual output.

Core Tech Stack

Component	Technology	Version
LLM Engine	Google Gemini 2.0 Flash	google-genai 1.64.0
Database	DuckDB (in-memory OLAP)	1.1.3
API Framework	FastAPI + Uvicorn	0.115.6
Frontend	Next.js + React + TypeScript	16.1.6 / 19.0.0
Voice STT	ElevenLabs Scribe v2	1.14.0
Voice TTS	ElevenLabs Flash v2.5	Streaming
Translation	Google Cloud Translate	3.18.0
Vector DB	ChromaDB + Sentence-Trans.	0.5.23
Charting	Recharts (5 chart types)	3.0.2
UI Library	Radix UI (28+ primitives)	Latest
Styling	Tailwind CSS 4	4.0
Animations	Framer Motion	12.23.24

2. PERFORMANCE & LATENCY METRICS

<2s

End-to-End Query

200ms

TTS First Byte

<500ms

Table Routing

Caching Strategy

- TTS Audio Cache: 500 MB max, 24-hour TTL, disk-based (eliminates repeat API calls)
- Schema Profile Cache: 3,600s TTL, in-memory (instant table metadata access)
- Query Result Cache: REMOVED by design (prevents stale/wrong data in multi-user scenarios)
- DuckDB Snapshots: Persistent .duckdb file (sub-100ms analytical queries)

Rate Limiting & Throughput

- 60 requests/minute per IP (sliding window, configurable via RATE_LIMIT_RPM)
- Exempt endpoints: /, /api/health, /api/auth/check
- HTTP 429 response with Retry-After header on violation

Query Execution Performance

Metric	Value	Notes
STT Transcription	<2 seconds	ElevenLabs Scribe v2
TTS First Audio Chunk	200-500ms	Streaming (vs 2-4s buffered)
LLM Planning	1-3 seconds	Gemini 2.0 Flash
SQL Compilation	<50ms	Template-based, no string interp
DuckDB Query	<100ms	In-memory OLAP engine
Self-Healing Retry	3 attempts	Fuzzy match threshold: 0.90
Max Result Set	10,000 rows	Configurable limit
Connection Pool	10 connections	DuckDB max connections
LLM Router Timeout	10 seconds	Falls back to rule-based
Request Timeout	20 seconds	Configurable in settings.yaml

3. NLP & VOICE CAPABILITIES

Bilingual NLP Engine (Tamil + English)

- Language auto-detection via langdetect (no user config needed)
- Google Cloud Translate API for Tamil <-> English translation
- 50+ entity extraction keywords (metrics, dimensions, identifiers, locations)
- 24 Tamil month name variations recognized (January -> Janvari)
- 31 Tamil ordinal number forms for date extraction
- Follow-up detection: pronouns, comparison phrases, time ranges
- Greeting detection: 26 Tamil starters + 22 English variations

Voice Activity Detection (VAD) -- Proprietary

Parameter	Value	Purpose
FFT Size	256-point	Frequency analysis
Smoothing Constant	0.3	Noise reduction
Silence Threshold	8 (audio level)	Speech vs silence
Min Speech Duration	500ms	Avoid premature cutoff
Silence Duration	1.5 seconds	Natural pause detection
Check Interval	50ms	Real-time responsiveness
Max Recording	15 seconds	Auto-stop timeout
Audio Codec	WebM/Opus	Bandwidth optimized

Voice UX Features

- Streaming TTS: Audio plays as chunks arrive (200ms TTFB, not 2-4s buffered)
- Live Captions: Word-by-word animation at 80ms intervals
- Always-On Mode: Continuous listening, auto-resumes after TTS finishes
- Farewell Detection: Recognizes goodbye phrases in both languages
- Haptic Feedback: Vibration patterns on record start/stop
- Dual Voice: Separate English and Tamil neural voices (ElevenLabs)
- Voice Visualizer: 32-bar frequency spectrum with ambient glow effects
- 1.15x Playback Speed: Optimized for natural conversation pacing

4. DATA INTEGRATION -- PLUG & PLAY

9 Data Source Connectors

Connector	Formats	Detection
Google Sheets	Sheets API	spreadsheets.google.com
Google Drive File	CSV, Excel, PDF	drive.google.com/file
Google Drive Folder	All files in folder	drive.google.com/folders
CSV	.csv	Extension / content-type
Excel	.xlsx, .xls, .xlsm	Extension detection
PDF	.pdf (table extraction)	pdfplumber library
Dropbox	CSV, Excel	dropbox.com URLs
OneDrive/SharePoint	CSV, Excel	sharepoint.com URLs
Local Filesystem	Any supported format	file:// or absolute paths

Auto-Profiling System

- Automatic table type detection: transactional, summary, item-level
- Column classification: metrics (50+ keywords), dimensions (20+), identifiers (25+)
- Date range inference per table (enables temporal filtering)
- Data quality scoring: 0-100% per table
- NULL analysis: warnings for >50% NULL columns
- Sample size: 10,000 rows for profiling analysis
- Entity learning: locations, categories, products extracted from data profiles

Data Sync & Change Detection

- SHA-256 hash-based change detection at sheet level
- Incremental rebuilds: only changed sheets re-processed
- Failure isolation: old data preserved if sync fails
- SSE loading progress: real-time phases (scanning -> profiling -> ready)
- Auto-load on startup: DEFAULT_DRIVE_FOLDER_URL env variable

Export Capabilities

- Excel export via XLSX library (client-side)
- PDF export via jsPDF with auto-table formatting
- COUNT queries show underlying data rows, not just the count

5. SECURITY & PRODUCTION HARDENING

Security Architecture

- SQL Injection Prevention: 18 regex patterns (UNION, xp_, WAITFOR, hex payloads)
- SSRF Prevention: Whitelist of 8 allowed domains (Google, Dropbox, OneDrive, etc.)
- Private IP Blocking: 127.x, 10.x, 172.16.x, 192.168.x all rejected
- Input Validation: 10,000 char max, NULL byte removal, quote escaping
- Auth: Token-based (64-char hex), SKIP_AUTH gated to dev environment only
- Debug endpoints: Protected behind require_auth middleware
- Prompt Injection Protection: 100 char limit on user memory inputs

Hallucination Prevention

- System prompt guardrail: 'ONLY use actual data numbers' (enforced 3x in prompts)
- No-speculation mode: 'Report what was searched factually, never analyze WHY'
- NaN/None handling: Skip invalid values instead of defaulting to 0
- Fuzzy match safety: 0.90 threshold prevents 'Oct' -> 'Nov' type errors
- Healing transparency: Surface all column/table fixes to user in response

Code Quality Metrics

Metric	Before	After
TypeScript `any` types	11	0
console.log statements	99	0
Python print() statements	550+	0 (structured logging)
ignoreBuildErrors	true	REMOVED
Silent except: pass	2	0 (all logged)
Dependency pinning	>=	== (exact versions)
services.py lines	4,176	1,285 + 6 modules
ChatScreen.tsx lines	2,860	981 + 5 components

CI/CD & Testing

- GitHub Actions: Automated on push to main/dev + PR validation
- Backend: pytest + coverage (30% minimum threshold)
- Frontend: TypeScript type check + ESLint + Next.js production build
- 155 test cases across 11 test modules
- Sentry integration: Optional error tracking via SENTRY_DSN env var
- Global exception handler: Returns JSON, never raw 500 HTML errors

6. WHAT MAKES THARA AI UNIQUE

Technical Differentiators vs Competitors

Cross-Table Intelligence

Multi-step query engine joins data across tables automatically. Example: 'Who worked on peak sales dates?' fetches sales peak date, then queries attendance -- no SQL knowledge needed.

Self-Healing Queries

If a query fails (typo, wrong column name, case mismatch), the engine automatically retries with 5+ fixing strategies. 0.90 fuzzy match threshold ensures accurate recovery without false positives.

True Bilingual -- Not Translation

Native Tamil NLP with 24 month names, 31 ordinal numbers, entity extraction in both languages. Not a translation wrapper -- understands Tamil grammar and context natively.

Voice-First Architecture

Proprietary VAD with 50ms polling. Streaming TTS delivers first audio in 200ms (vs 2-4s industry standard). Always-on mode enables hands-free continuous conversation.

Zero-Config Data Onboarding

Drop a Google Drive folder link -- auto-loads all files, profiles every column, detects date ranges, classifies table types. Zero schema definition required.

Personality-Driven UX

Named assistant 'Thara' with 50+ natural conversation patterns in both languages. Emotion detection adapts response tone. Never asks 'Did you mean?' -- makes confident decisions.

Enterprise Security

18-pattern SQL injection defense, SSRF whitelist, private IP blocking, rate limiting, prompt injection protection. Production-ready from day one.

Dataset Agnostic

Works with ANY domain -- sales, HR, healthcare, education, logistics. All routing logic derives from loaded data profiles, zero hardcoded domain knowledge.

THARA AI = Voice + Data + Intelligence

Talk to your business data in your language. Get instant, accurate, trustworthy answers.

No SQL. No dashboards. No training. Just ask.

